2. Row 1 Leading element $= 2$ in Column 1
Row 2 Leading element $= 2$ in Column 2
$\therefore$ The condition that for each row, the leading element is in a column to the right of all such columns of the leading elements of all rows above it is True. $\square$

3. Row 2 Column $1 = 0$. $\therefore$ The condition that for each pivot column, all column elements below the pivot position is 0 is True.
$\therefore$ The matrix $\begin{bmatrix} 2 & 2 & h \\ 0 & 2 & k+h/2 \end{bmatrix}$ is in echelon form.

There are no rows in the matrix of the form $[0\ 0\ 0 \ldots 0\ b]$, where $b \neq 0$. By Theorem 2, the given matrix has a system which is consistent, ie. atleast one solution set exists.
$\therefore \begin{bmatrix} h \\ k \end{bmatrix}$, for any arbitrary h, k is a linear combination of $u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. $\therefore \begin{bmatrix} h \\ k \end{bmatrix}$ is in the span of $\{u, v\}$ $\forall h, k \in \mathbb{R}$. $\square$

22) Construct a 3×3 matrix A, with nonzero entries, and a vector b in $\mathbb{R}^3$ such that b is not in the set spanned by the columns of A.

$\boxed{\text{Solution:}}$ $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 5 & 4 \\ 3 & 7 & 3 \end{bmatrix}$ $b = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}$

$\boxed{\text{Claim:}}$ The vector b is not in the set spanned by the columns of A.

$\boxed{\text{Proof:}}$ Since b is not in the set spanned by the columns of A, by defn. of span, $\forall x_1, x_2, x_3 \in \mathbb{R}$, $x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 4 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}$.

This is equivalent to claiming that the augmented matrix $\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 5 & 4 & 2 \\ 3 & 7 & 3 & 3 \end{bmatrix}$ has no solution set. We will verify that